Exploring Large Language Models for Pathogen Genomic Metadata Extraction

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Abstract:

The lack of standardized pathogen sequence metadata poses significant challenges for interoperability and use in public health research. Currently, due to the various formats of information spread across lab reports, publications, and datasets, manually compiling relevant data is time intensive, costly, and inefficient with the emergence of new pathogens. The aim of this project is to develop a Large Language Model (LLM) pipeline to automate the extraction and harmonization of metadata from unstructured text sources to develop structured pathogen metadata repositories. In this project we created a reference dataset of studies containing respiratory virus metadata, which will be used to evaluate the LLM pipeline for accuracy and completeness. To test the performance of the LLMs, prompts will be iteratively engineered to achieve optimal extraction of metadata from each study. This will allow for consistent access to pathogen sequence metadata for public health laboratories across the United States.

Introduction:

- Genome sequencing has drastically improved public health by enabling the monitoring, characterization, and identification of emerging pathogens.
- Metadata, data that describes data, is essential for supporting accurate analysis, understanding genomic information, and informing public health decision-making
- Advances in LLMs present an opportunity to automate and standardize metadata extraction.
- This project aims to develop an LLM-based pipeline that automates the curation and harmonization of pathogen sequence metadata to populate pathogen metadata repositories.

Methods:

1. **Reference dataset:** Contains the BioProject accession ID, Organization and Species (SARS-CoV-2, Influenza, Respiratory Syncytial Virus, or Candida Auris) for a total of 49 pathogen sequences
2. **LLM Comparison:** Determine the performance metrics of metadata extraction between different LLMs (Gemini, Perplexity, Claude, ChatGPT 4o mini, and Grok)
3. **Prompt Engineering**: Optimal zero-shot prompt for LLM query
4. **Gold Standard**: Create a dataset of ground truth metadata used for extraction evaluation
5. **Evaluate:** Ensure standardization, quantitative and qualitative performance

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Implementing LLMs for pathogen metadata extraction can reduce processing time by at least 98% compared to manual curation.

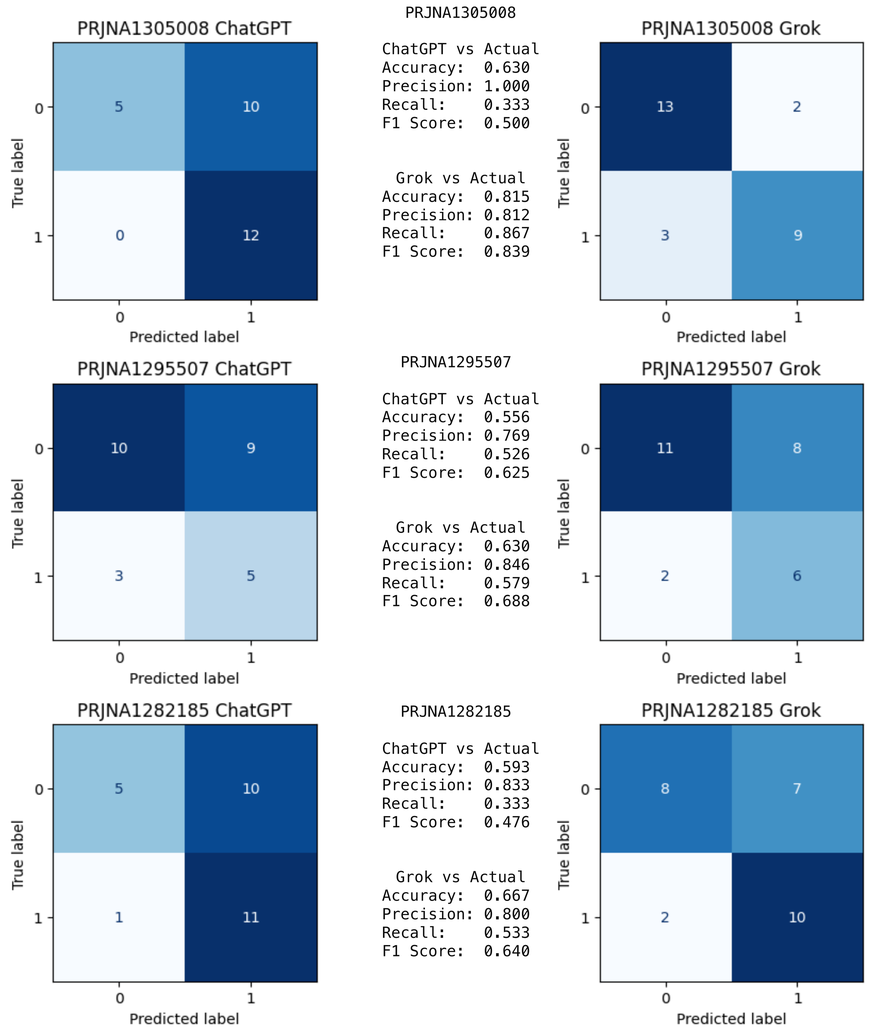

Figure 1: Confusion matrix of three different pathogens for ChatGPT and Grok

Results:

Gemini and Claude could not access the BioSample webpage link, while Perplexity could not display the information. Because of the inability to access and display the necessary attributes, Gemini, Claude, and Perplexity were not included in the remainder of comparison metrics.

Time of Manual vs Automated Metadata Extraction:

- Manual extraction: ~ 2400 Seconds (40 Minutes per paper)
- Automated extraction
    - Grok: ~37 Seconds
    - ChatGPT 4o: ~7 Seconds

Automated extraction resulted in at least a 98.53% reduction in time

Accuracy of Automated Metadata Extraction:

- Grok Average Accuracy = 0.704
- ChatGPT Average Accuracy = 0.593

Grok on average performed better than ChatGPT across all metrics aside from precision

Discussion:

While general metadata, such as the species and institution name is consistently available, more sample or sequence specific metadata is excluded. Although the completeness of metadata is rather inconsistent (figure 2), both LLMs perform relatively well across all metrics, especially considering the drastic decrease of time it takes to curate the information.

Next steps are to perform prompt engineering to ensure an optimal zero-shot prompt for extracting metadata from various sources. Additionally, to use standard information retrieval evaluation methods with two independent reviewers extracting metadata manually and a third comparing the results of the two independent reviewers to create a gold standard. The gold standard will act as ground truth for LLM performance evaluation.

Extract the metadata for the following attributes from this biosample record and create a table with each attribute as a row: 'Bioproject Link'

coordinate (x, y, z), Street 1 address, Street 2 address, State, City, Zip code, Country, Institution name, Sequence run ID, Sequence run date, Platform name, Platform version, Primer sequence name, Primer sequence version, Primer sequence, Taxonomy database name, Taxonomy database level, Taxonomy database ID, Lineage nomenclature, Lineage nomenclature ID, String-based representation, Host sex, Host age (or age bin), Host ethnicity, Sample submission date and time, Sample processing date and time, Sample collection date and time.

You are welcome to use any sources that will help you find as many of the values as possible. Some you may find helpful will be biosample, NCBI taxonomy, and even any published work that may be relevant.

Attribute	Coverage (%)
Country	100.000000
Taxonomy database name	100.000000
Taxonomy database level	100.000000
Taxonomy database ID	100.000000
Sequence run ID	100.000000
Platform name	100.000000
Institution name	85.714286
State	85.714286
City	71.428571
Sample collection date and time	71.428571
Sample submission date and time	57.142857
Zip code	57.142857
Platform version	57.142857
Primer sequence name	42.857143
Lineage nomenclature ID	42.857143
Lineage nomenclature	42.857143
Street 1 address	42.857143
String-based representation	42.857143
Coordinate (x, y, z)	28.571429
Sequence run date	28.571429
Primer sequence	14.285714
Street 2 address	14.285714
Host age (or age bin)	14.285714
Sample processing date and time	0.000000
Host sex	0.000000
Host ethnicity	0.000000
Primer sequence version	0.000000

Figure 2: Example input into the LLM (left) and coverage of metadata across seven of the reference pathogen sequences

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